Total No. of Questions : 10]

SEAT No. :

P3052

[Total No. of Pages : 4

[5670] - 151

B.E. (Mechanical Engineering) (Semester - II)

FINITE ELEMENT ANALYSIS

(2012 Pattern) (Elective - IV)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates :

1) *Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8, Q.9 or Q.10.*
2) *Draw suitable neat diagram, wherever necessary.*
3) *Figures to the right indicate full marks.*
4) *Use of electronic pocket calculator is allowed.*
5) *Assume suitable data, if required.*

Q1) a) State different approaches used in FEA. Write a note on "Principle of Minimum potential energy (PMPE)". **[6]**

b) State and explain different steps in FEM. **[4]**

OR

Q2) 2 For truss given in figure 1, determine : **[10]**

a) Nodal displacements.

b) Stresses in each element and

c) Reaction at support. Take E = 200 x 10^3N/mm^2 and A = 1500 mm^2.

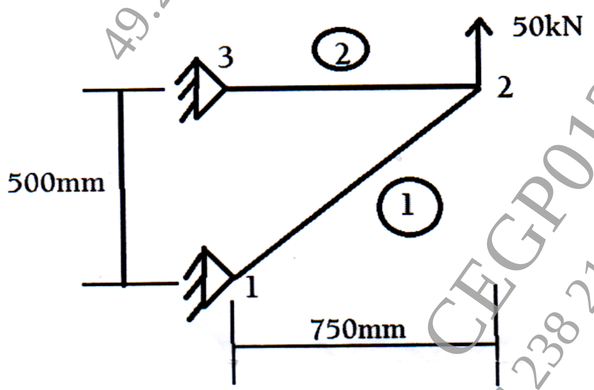

Figure:1

P.T.O.

Q3) a) Explain strain - displacement relationship matrix for CST element. **[6]**

b) Explain how the different types of continuities are important in FEA during element formulation. Explain C^0 and C^1 type of continuity. **[4]**

OR

Q4) a) Determine x and y coordinate of the interior point P for the CST element as shown in figure 2, Using co-factor method. A CST element is defined by nodes, I (1, 1), J (6, 3) and K (4, 5). The shape functions are N1 = 0.2 and N2 = 0.3. **[4]**

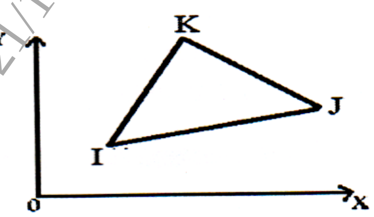

Figure:2

b) Show that skyline matrix method is useful for reduction in memory required for simulation in FEM. **[6]**

Q5) a) Explain following : **[8]**

i) Higher order elements : Lagrangean and Serendipity Elements.

ii) Different types of Quality checks for meshing in FEM.

b) For four noded quadrilateral element as shown in figure 3, assemble Jacobin matrix and shape functions for the Gauss point (0.7, 0.5). **[8]**

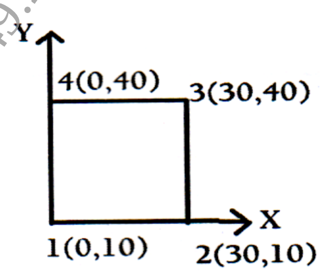

Figure:3

OR

Q6) a) Explain Lagrange interpolation function used in FEA. **[8]**

b) Evaluate following Integral using 2-point and 3-point Gauss Quadrature method **[8]**

$$I=\int_{-1}^{1}\left[1+x+x^{2}\right]dx$$

Q7) a) Write Element thermal conductivity (stiffness) matrix and Element thermal load vector. Compare them with Element stiffness matrix and element load vector of Bar element respectively. **[8]**

b) A brick wall has thickness of 0.6m and thermal conductivity of 0.75 W/m°K. The inner surface of wall is at 15°C and outer surface is exposed to cold air at –15°C. The convection heat transfer coefficient at outer surface is 40W/m5°K. Using two elements for finite element formulation, determine the steady state temperature distribution within the wall and heat flux through the wall. **[10]**

OR

Q8) a) State and briefly explain steps to solve 1-D steady state heat transfer problem using finite element method. **[6]**

b) The fin shown in figure 4, is insulated on the perimeter. The left end has a constant temperature of 100°C. A positive heat flux of q = 5000W/m^2 acts on the right end. Let k = 6w/m°C and cross sectional area A = 0.1 m^2. Determine the temperature at 0.25 L, 0.5L, 0.75L and L, where L = 0.4 m. **[12]**

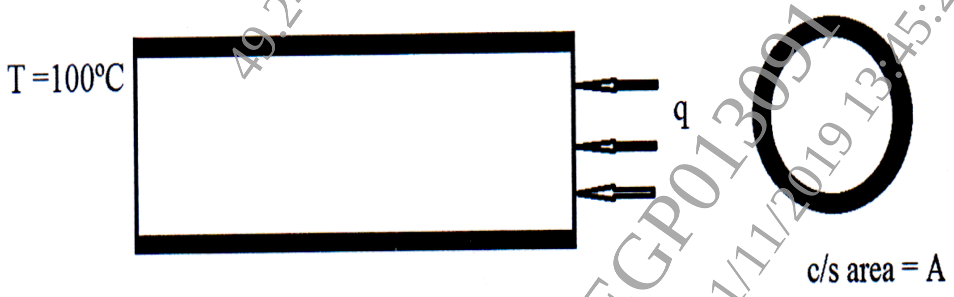

Figure 4

Q9) a) For undamped-free vibration system, explain Eigen-value problem. **[6]**

b) Using consistent mass matrix method, find the natural frequency of longitudinal vibration with two elements of bar as shown in figure 5. Take $E = 2 \times 10^{11}$ N/m^2, $\rho = 7800$kg/m^3, L = 1m, $A = 30 \times 10^{-6}$m^2. **[10]**

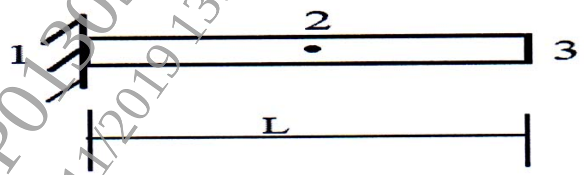

Figure:5

OR

Q10) a) Use consistent mass matrix and find the first two natural frequencies of longitudinal vibration of the stepped bar as shown in figure 6. Take cross-sectional areas A1 = A3 = 5cm^2 and A2 = 10 cm^2, E = 210 GPa and ρ = 7860 kg/m^3. **[10]**

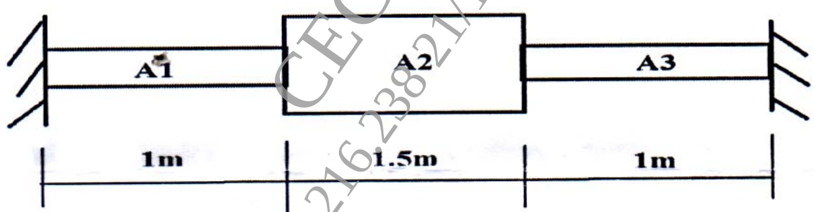

Figure:6

b) Write consistent and lumped mass matrix for **[6]**

i) Bar element.

ii) Truss element and

iii) Beam element.